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IRS PROJECT PROPOSAL

Date of proposal: 13/09/2026

Project Title: OptEV+ | Intelligent EV Charging Infrastructure Planner

Group ID: 25

Group Members:

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Background

Singapore aims to deploy 60,000 electric vehicle charging points by 2030: 40,000 in public car parks and 20,000 in private premises. By September 2026 around 34,700 were in place, serving an EV population that had passed 82,100 in August. Every HDB town has been declared EV-ready, more than 90% of HDB car parks now have charging points, and every town has at least one fast charging hub by the end of 2027.

The next phase of deployment opens a richer planning problem. Capacity will come from a wider range of locations — HDB and public car parks, petrol stations, hawker centers, community and commercial facilities, and existing charging sites with room to grow. Across these, planners decide not only which sites qualify, but how much capacity each one needs and how a limited budget is best distributed across a large pool of candidates.

Public data can help answer these questions, but it requires preparation and interpretation. Charger status and car park availability are published through separate feeds. They must be collected over time, matched geographically, and assessed for missing or outdated records before they can support planning decisions.

Utilization alone is also insufficient. A busy site may have no room or electrical capacity for additional chargers, while a site without chargers produces no utilization data at all. Recommendations therefore need to account for demand, existing supply, and site constraints together. They should also retain the evidence and reasons behind each decision so that planners can review and explain the proposed allocation.

Objective

To develop an intelligent reasoning system that supports government planners in deciding where additional public EV charging points should be deployed and how many are required at each location, producing a ranked set of candidate sites, a constrained allocation for a given budget and quantity, and a traceable justification for every site accepted, rejected or deferred.

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Scope

The data collection is island wide. Singapore has 55 planning areas, and analysis is aggregated at that level. Detailed evaluation and the demonstration will focus on one planning area, selected after the first week of collected data against a criterion stated in advance.

Out of scope: commercial tendering, consumer-facing charger recommendation, electrical engineering approval, and any use of vehicle owner records. Recommended locations are planning candidates. Electrical capacity and land access require separate verification before construction.

Proposed System

Data foundation. Scheduled collectors will gather charger status and car park availability across Singapore. These observations will be combined with car park records, planning area boundaries, and candidate amenities. The pipeline will standardize coordinates, align timestamps, match locations at several distance thresholds, and assign records to planning areas. Data coverage and freshness will be reported alongside the analysis.

Measurement and inference. Accumulated observations will be used to derive utilization, saturation, and charging-pressure indicators. Occupancy indicates how much available charging capacity is being used but does not directly represent overall charging demand. The system will therefore infer demand from the available observations and supporting location characteristics, while reporting the reliability and limitations of these estimates. Where a site is consistently saturated, the data will be treated as supporting a lower-bound estimate of demand rather than a precise value. Sites without existing chargers provide no direct occupancy signal, so their potential demand will be estimated using comparable locations and available contextual data.

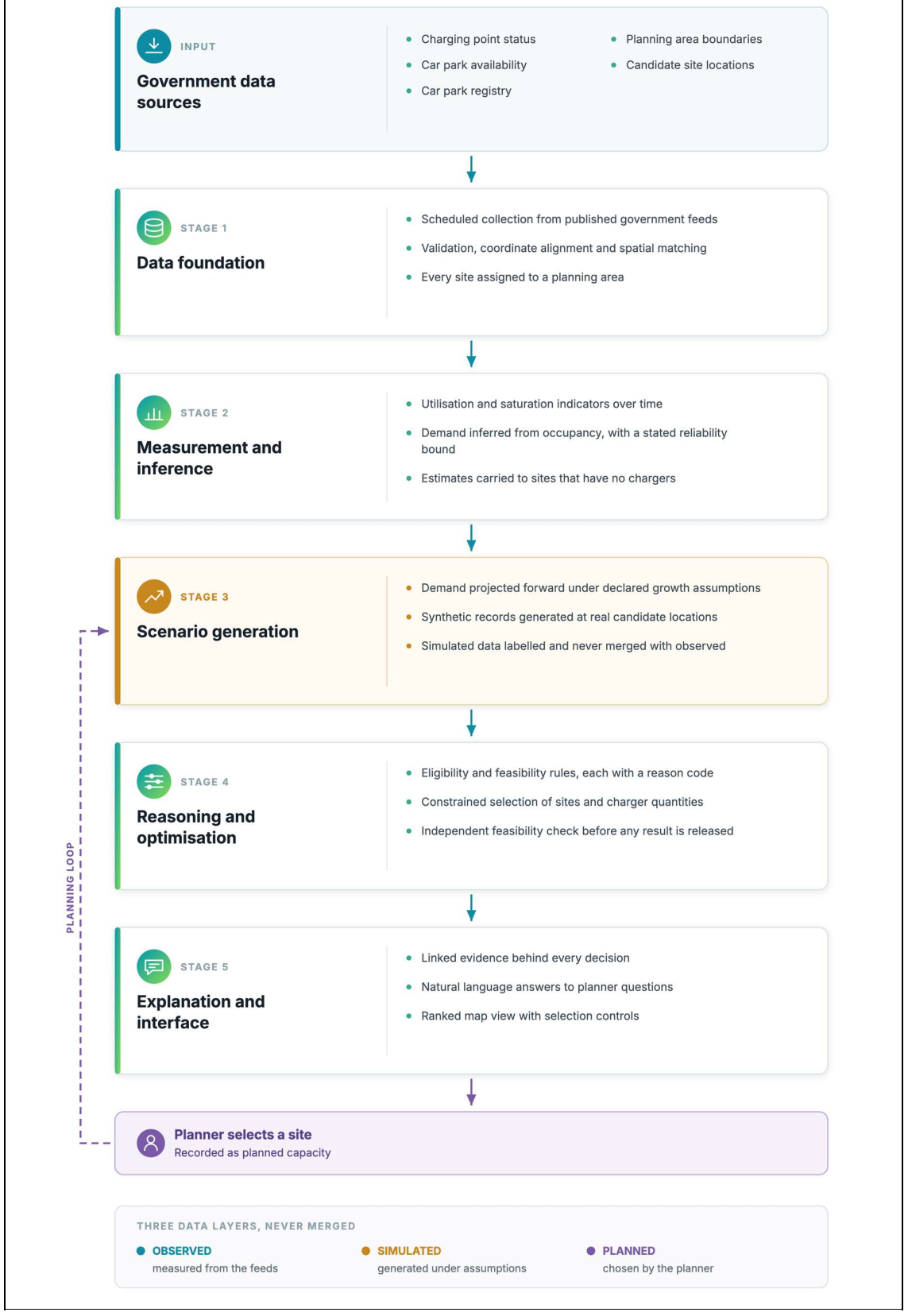
Reasoning and optimization. Rules will assess candidate eligibility and feasibility, recording a reason for each outcome. The optimizer will then allocate charging points subject to budget, site capacity, coverage requirements, and policy constraints. A separate checker will re-evaluate every hard constraint before a plan is published. If the requested allocation cannot be satisfied, the system will report on the unmet requirement and the relevant constraints.

Explanation and interface. A knowledge graph will connect sites, observations, indicators, rules, assumptions, and planning decisions. A natural-language interface will retrieve this evidence to explain recommendations and answer planner questions. The dashboard will display candidate coordinates, rankings, and allocations on a map. Selected sites will be recorded as planned capacity, separately from operational infrastructure and observed data. Planners can therefore revise their choices and recalculate a scenario without changing the underlying measurements.

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System Design



Data Sources and Known Limitations

All primary data comes from Singapore government sources:

- Charging point status and car park availability, both published as live feeds
- A static car park registry giving location and site characteristics
- Planning area boundaries for geographic aggregation
- Locations of public amenities used as additional candidate sites

Access methods, refresh rates, field coverage, and record counts will be confirmed in Week 1, before the collection design is fixed. Several known limitations already shape the design and will be quantified during that assessment. Live availability feeds report a level rather than a flow, so individual vehicle movements cannot be recovered. Charger status codes group several conditions together, so some records must be filtered before capacity is counted. Identifiers do not align perfectly across sources, so joints are partly spatial and their coverage must be reported rather than assumed. Vehicle-level records are not publicly available and are not used anywhere in this design. Demand beyond the observation period is generated under declared assumptions and kept separate from measured data at every stage.

Technique Group Mapping

The system demonstrates all four technique groups, and each one feeds the next.

Decision automation. Eligibility and feasibility rules evaluated over structured facts, producing an accept, reject, defer or conditional outcome with the reason recorded alongside it.

Business resource optimization. Constrained selection of sites and charging point quantities under budget, capacity and coverage limits, framed as a search over candidate allocations and checked before any result is released.

Knowledge discovery and data mining. Discovery of demand patterns from observed occupancy, characterizing how usage varies by location and time of day, and a model that carries demand estimates across sites with no charging supply.

Cognitive techniques. A knowledge graph linking sites, evidence, rules, assumptions and decisions, supporting queries across those links and a natural language explanation layer for government users.

Discovered patterns inform the demand model. The demand model produces the quantities the allocation works with. The rules are bound to what it may select. The graph explains the results.

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Timeline

PROPOSAL DUE 13 SEP — **FINAL SUBMISSION 25 OCT**

W0	W1	W2	W3	W4	W5	W6
12 – 13 Sep	14 – 20 Sep	21 – 27 Sep	28 Sep – 4 Oct	5 – 11 Oct	12 – 18 Oct	19 – 25 Oct

OBSERVED live collection

SIMULATED Generated from the week 1 baseline, labelled and kept separate

01 W0 – W1 · 12 – 20 Sep

Proposal and setup

- Proposal submitted and presented
- One week of live data collected
- Interfaces frozen so the team works in parallel

02 W2 – W3 · 21 Sep – 4 Oct

Data foundation

- Baseline built from the collected week
- Utilisation and saturation indicators
- Demand patterns across sites and time

03 W4 – W5 · 5 – 18 Oct

Reasoning and optimisation

- Eligibility rules and constrained allocation
- Evidence and explanation layer
- Feature freeze on 18 Oct, experiments run

04 W6 · 19 – 25 Oct

Evaluation and delivery

- Comparison against baselines under equal constraints
- Report, two videos and user guide
- Final submission on 25 Oct

FINAL DELIVERABLES

- Runnable intelligent reasoning system
- Two presentation videos
- Individual reflections and peer review
- Group project report with appendices
- Installation and user guide
- Project repository

NUS National University of Singapore | ISS

References

Policy and regulations:

- Land Transport Authority, Our EV Roadmap. https://www.lta.gov.sg/content/ltagov/en/industry_innovations/technologies/electric_vehicles/our_ev_roadmap.html
- Land Transport Authority, Factsheet: Accelerating Nationwide Deployment of Electric Vehicle Charging Points. https://www.lta.gov.sg/content/ltagov/en/newsroom/2021/3/news-release/Accelerating_nationwide_deployment_of_electric_vehicle_charging_points.html
- Land Transport Authority, Regulating the Safety of EV Chargers. https://www.lta.gov.sg/content/ltagov/en/industry_innovations/technologies/electric_vehicles/regulating_safety_of_ev_chargers.html
- Ministry of Transport, Electric Vehicles. https://www.mot.gov.sg/what-we-do/green-transport/electric-vehicles/
- Ministry of Transport, Deployment of EV Charging Infrastructure in New Towns. https://www.mot.gov.sg/news-resources/newsroom/deployment-of-ev-charging-infrastructure-in-new-towns/
- Charging point and electric vehicle population figures as reported in September 2026. https://www.asiaone.com/singapore/electric-vehicle-charging-points-singapore-number-grows

Data sources:

- LTA DataMall, developer portal. Electric vehicle charging point status and car park availability. https://datamall.lta.gov.sg/content/datamall/en.html
- LTA DataMall, API User Guide and Documentation. https://datamall.lta.gov.sg/content/dam/datamall/datasets/LTA_DataMall_API_User_Guide.pdf
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- Urban Redevelopment Authority, Master Plan 2025 Planning Area Boundary. https://data.gov.sg/datasets/d_2cc750190544007400b2cfd5d7f53209/view
- Urban Redevelopment Authority, Master Plan 2025 Land Use Layer. https://data.gov.sg/datasets/d_a8c3546b26712e35021f3a681d0353ae/view
- National Environment Agency, Hawker Centres. https://data.gov.sg/datasets/d_4a086da0a5553be1d89383cd90d07ecd/view

AI Declaration

In the preparation of this proposal, AI-based tools including Claude, ChatGPT, and Gemini were used in a supporting capacity to assist with tasks such as drafting and refining text, organizing content, and providing general research and reference support. All ideas, analysis, technical design decisions, and conclusions presented in this proposal are the work of the project team, with AI tools used solely to support and streamline the writing and preparation process.